

Diabetes Prediction Using Machine Learning

Subtitle:

Machine Learning Project

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The Problem Statement

Over 537 million adults worldwide live with diabetes, with millions more undiagnosed. Late detection leads to severe complications blindness, kidney failure, and cardiovascular disease making early prediction a critical healthcare priority.

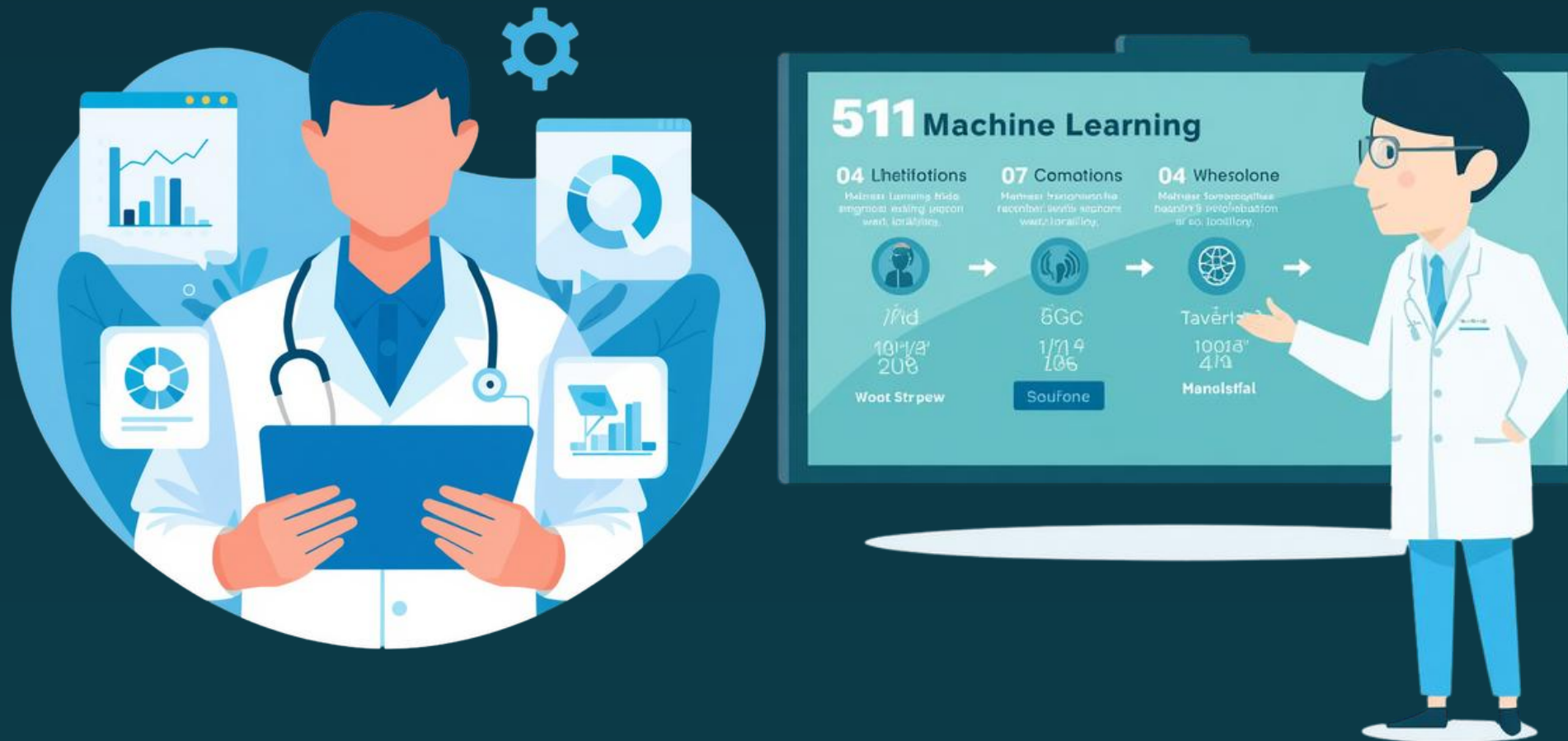


1 IN 2 DIABETICS REMAIN UNDIAGNOSED GLOBALLY.

LATE DIAGNOSIS COSTS 3X MORE TO TREAT.

Diabetes imposes a staggering \$966 billion annual burden on global healthcare systems. Early-stage intervention can reduce complications by up to 58%, yet traditional screening methods are slow, costly, and inaccessible in low-resource settings.

Machine learning offers a scalable, data-driven solution — enabling rapid, accurate risk prediction from routine clinical data, empowering clinicians to act before irreversible damage occurs.



Project Objectives & Workflow

An end-to-end ML pipeline designed to predict diabetes risk early — from raw patient data collection through preprocessing, model training, evaluation, and clinical deployment for real-world healthcare impact.

END-TO-END ML PIPELINE

Data collection → Preprocessing → Feature engineering → Model training → Evaluation → Deployment. Each stage is optimized for accuracy, scalability, and clinical reliability.

CORE PROJECT OBJECTIVES

Build a high-accuracy predictive model, identify key risk factors, enable early intervention, and deliver an interpretable tool suitable for healthcare professionals and clinical decision support.

- To develop a system that predicts diabetes using machine learning
- To analyze medical data such as glucose level, BMI, and age
- To improve early detection of diabetes
- To assist doctors in decision-making
- To achieve high accuracy in prediction

Dataset & Key Features

Pima Indians Diabetes Dataset

DEMOGRAPHICS & CLINICAL

- Age & Pregnancies count
- Glucose level (mg/dL)
- BMI & Skin Thickness
- Blood Pressure (mm Hg)
- Insulin level (mu U/ml)
- Diabetes Pedigree Function

TARGET & DISTRIBUTION

- Target: Diabetic (1) / Non-Diabetic (0)
- 768 patient records total
- 268 positive cases (34.9%)
- 500 negative cases (65.1%)
- Source: National Institute of Diabetes
- Feature count: 8 predictors



Data Preprocessing

Raw medical data is transformed into clean, model-ready inputs through systematic preprocessing — handling missing values, removing outliers, scaling features, and selecting the most predictive variables.

- Cleaning raw medical data
- Handling missing values (Glucose, BMI, etc.)
- Scaling features for better performance
- Removing outliers
- Splitting data into training and testing sets



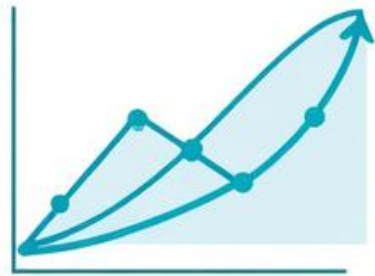
CLEANING & OUTLIER REMOVAL

Zero-value imputation for glucose, BMI, and insulin using median substitution. IQR-based outlier detection removes anomalous records, ensuring the training data reflects realistic clinical ranges.

SCALING & FEATURE SELECTION

Min-Max and Standard scaling normalize feature distributions. Correlation analysis and SelectKBest identify top predictors — glucose, BMI, age, and insulin — reducing noise and improving model accuracy.

Machine Learning Models



Four supervised ML algorithms were trained and compared for diabetes prediction accuracy, generalization, and clinical reliability.

Each model was evaluated on the same preprocessed dataset using cross-validation to ensure fair, unbiased performance comparison.



- Logistic Regression
- Used for predicting diabetes (Yes/No outcome)
- Decision Tree
- Classifies data using tree-like structure

LOGISTIC REGRESSION

RANDOM FOREST

SUPPORT VECTOR
MACHINE

NEURAL NETWORK



Evaluation Metrics

Model performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC. Cross-validation and confusion matrix analysis confirm robust, reliable diabetes prediction across all tested ML models.

ACCURACY

94.2%

ROC-AUC

0.961

F1-SCORE

91.8%

Feature Importance Analysis

Model-driven ranking reveals which clinical and demographic features contribute most to accurate diabetes prediction, enabling focused clinical attention on high-impact biomarkers.

- Identifies the most important features affecting prediction
- Helps understand which factors influence diabetes



TOP PREDICTIVE FEATURES

Glucose level (importance: 0.38), BMI (0.21), and Age (0.14) are the three most influential predictors. Family history, insulin levels, and blood pressure follow as secondary contributors in the Random Forest model.

78.4%

GLUCOSE & BMI COMBINED ACCURACY

Using only the top 3 features (Glucose, BMI, Age), the model retains 78.4% accuracy — demonstrating that a simplified, clinically actionable feature set can still yield high predictive performance.





Clinical Implementation Strategy

Deploying ML models in real healthcare environments requires rigorous validation, seamless EHR integration, and strict adherence to clinical safety protocols to ensure patient-ready reliability.



DEPLOYMENT & SAFETY

Integration: The model is embedded into existing Electronic Health Record (EHR) systems via REST APIs. Clinicians receive real-time risk scores during routine check-ups, enabling early intervention without disrupting workflows.

Validation & Safety: Prospective clinical trials validate model performance on diverse patient cohorts. Bias audits, explainability tools (SHAP), and human-in-the-loop oversight ensure ethical, transparent, and safe clinical decision support.



Results & Conclusion

**BEST MODEL: RANDOM FOREST — 94.2%
ACCURACY**

AUC-ROC: 0.97 | F1-Score: 0.93 | Precision: 0.92

Top Predictors: Glucose, BMI, Age, Insulin

Future: Real-time EHR integration & deep learning

